

PLASMA SPRAY PROCESSORS SOP for Inspection, Handling, and Verification of Thermal Coating Powder	QF-QMS/02/42	
	Rev. 00	
	Date : 15.04.2025	

Departments Involved: Stores / Quality Assurance (QA) / Laboratory

1. Purpose

To establish a standard procedure for receiving, inspecting, handling, and verifying thermal coating powder before it is approved for use in production.

2. Scope

This SOP applies to all batches of thermal coating powder received at the facility intended for use in thermal spray coating processes.

3. Responsibility

- **Store Personnel:** Material receipt and initial verification.
- **QA Department:** Inspection, sampling, documentation, and coordination of testing.
- **Lab Personnel:** Execution of test spray and metallographic analysis.
- **Spray Operator:** Conduct sample spraying using approved parameters.

4. Procedure

4.1 Material Receiving and Initial Check (Stores)

- Receive the thermal coating powder from the supplier.
- Visually inspect packaging for damage or leakage.
- Verify details against the packing list:
 - Supplier Name
 - Powder Name
 - Quantity
 - Batch/Lot Number
 - Date of Manufacture
- Place the material in the designated **“Under Inspection”** area.
- Notify the QA department for further inspection.

4.2 Quality Assurance (QA) Inspection

- QA logs the material in the **Raw Material Inspection Register**, recording:
 - Date of Receipt

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Verified By	Prasanna Wadkar	Signature		Date	

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○ Powder Name ○ Batch/Lot Number ○ Quantity ○ Certificate of Analysis/Test Certificate verification ● Collect one (1) box of powder from the received batch for validation/testing. 4.3 Sample Testing and Spraying ● QA hands over the sample box and approved spray parameters to the spraying operator. ● Perform test spray and deposit efficiency evaluation on standard test coupons. ● Ensure the correct spray parameters are followed, such as: ○ Spray Distance ○ Temperature ○ Powder Feed Rate ○ Gas Flows (Argon, Hydrogen, Nitrogen, Fuel) ○ Electrical Parameters (Amp, Volt) ○ Combustion Pressure 4.4 Metallurgical Analysis (Lab) ● Conduct metallographic examination on the sprayed sample: ○ Coating Hardness ○ Porosity Level ○ Oxide content ○ Microstructure Examination 4.5 Result Evaluation and Approval ● Compare test results with internal specifications or industry standards. ● If results meet acceptance criteria, QA approves the batch for use and moves it to the “Approved for Use” area. ● If the material fails, isolate and quarantine it for further action (re-test, & change parameter to achieve results)

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5. Documentation

- Raw Material Inspection Register
- Material Test Certificates
- Spray Test Results
- Metallographic Analysis Reports
- Approval/Rejection Record

6. References

- Internal Material Specification Sheets
- Thermal Spray Process Standard Procedure
- ASTM/ISO Testing Standards

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